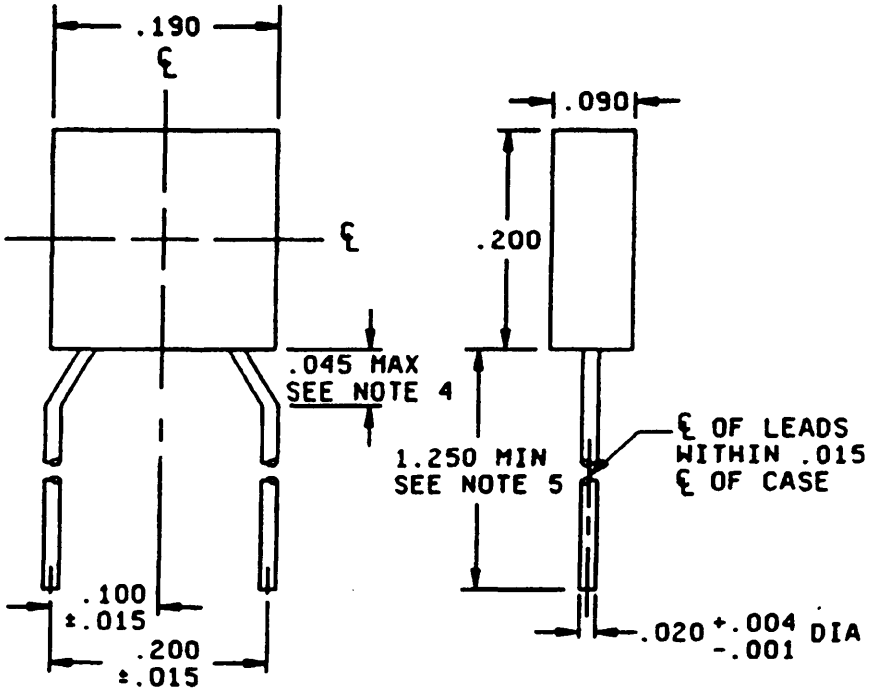
INCH-POUND
MIL-C-55514/8E
8 July 1993
SUPERSEDING
MIL-C-55514/8D
15 February 1989

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, METALLIZED PLASTIC DIELECTRIC, DC, IN NONMETAL CASES, ESTABLISHED RELIABILITY STYLES CFR12 AND CFR16

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55514.



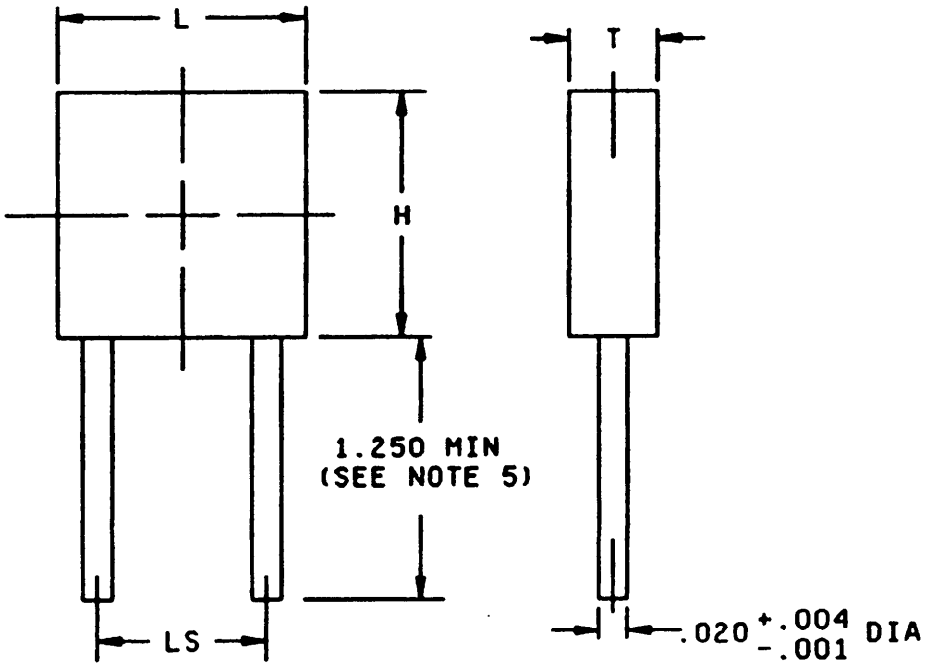
Style CFR12

Ⓔ FIGURE 1. Dimensions and configuration.

Ⓔ denotes changes

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Style CFR16

Inches	mm	Inches	mm
.001	0.03	.090	2.29
.004	0.10	.100	2.54
.015	0.38	.190	4.83
.025	0.64	.200	5.08
.045	1.14	1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. For flush mounting .078 (1.98 mm) printed-circuit hole diameter is required to clear shoulder.
5. Lead length may be a minimum of one inch long for use in tape and reel packaging, when specified in the ordering data.

(E) FIGURE 1. Configuration and dimensions - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Case type: Molded or preformed.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic R: Will operate to +125°C when properly derated to 50 percent of its +85°C rated voltage.

Voltage rating: See table I.

Dissipation factor: Not more than 0.3 percent.

(E) Preconditioning: Not applicable.

(E) Burn-in: DC: 48 hours at 125°C, at 1.4 times the derated 125°C voltage rating. The current to each capacitor shall be limited to 1 ampere maximum by a series resistor.

AC: Not applicable.

(E) Dielectric withstanding voltage: 200 percent of rated voltage for 10 to 60 seconds. The surge current shall be limited to 1 ampere, maximum.

(E) Resistance to soldering heat: In accordance with MIL-C-55514 except that spacers or standoffs of .25 inch are required.

Application note: These capacitors are heat sensitive. Vapor phase soldering is not recommended. For applications that require wave soldering that exceeds a 5 second duration, it is recommended that a spacer or standoff be used.

Marking: In accordance with MIL-C-55514 except that minimum marking as shown in the following example will be permitted. Full marking shall be included on the package.

	Front	Reverse side	
Type designation	CFR 12 RRA 333 FM	J-- 86 11	J = JAN, and two digit manufacturer's code or symbol. Date code in accordance with MIL-C-55514.

NOTE: Location of manufacturer's code or symbol is optional.

Substitutability for failure rate levels shall be as follows:

Failure rate level	Will replace failure rate level
S	R, P, M
R	P, M
P	M
M	

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TABLE I. Styles CFR12 and CFR16 capacitor characteristics.

Characteristic R capacitors											
Style CFR12 1/				Style CFR16							
Type designation 2/	Capacitance	Rated voltage @ +85°C	Capacitance tolerance	Type designation 2/	Capacitance	Rated voltage @ +85°C	Capacitance tolerance	T ±.010 (0.25)	N ±.020 (0.51)	L ±.010 (0.25)	L.S. ±.015 (0.38)
	µF	(volts)			µF	(volts)					
CFR12RRA333--	.033	50	F, G, J, K	CFR16RRA683--	.068	50	F, G, J, K	.095(2.41)	.245(6.22)	.295(7.49)	.200(5.08)
CFR12RRA393--	.039	50	F, G, J, K	CFR16RRA823--	.082	50	F, G, J, K	.095(2.41)	.245(6.22)	.295(7.49)	.200(5.08)
CFR12RRA473--	.047	50	F, G, J, K	CFR16RRA104--	.10	50	F, G, J, K	.095(2.41)	.245(6.22)	.295(7.49)	.200(5.08)
CFR12RRA563--	.056	50	F, G, J, K	CFR16RRA124--	.12	50	F, G, J, K	.095(2.41)	.295(7.49)	.295(7.49)	.200(5.08)
CFR12RRG223--	.022	75	F, G, J, K	CFR16RRA154--	.15	50	F, G, J, K	.095(2.41)	.295(7.49)	.295(7.49)	.200(5.08)
CFR12RRG273--	.027	75	F, G, J, K	CFR16RRA184--	.18	50	F, G, J, K	.095(2.41)	.295(7.49)	.295(7.49)	.200(5.08)
CFR12RRB103--	.010	100	F, G, J, K	CFR16RRA224--	.22	50	F, G, J, K	.095(2.41)	.395(10.03)	.395(10.03)	.300(7.62)
CFR12RRB123--	.012	100	F, G, J, K	CFR16RRA274--	.27	50	F, G, J, K	.095(2.41)	.395(10.03)	.395(10.03)	.300(7.62)
CFR12RRB153--	.015	100	F, G, J, K	CFR16RRA334--	.33	50	F, G, J, K	.095(2.41)	.395(10.03)	.395(10.03)	.300(7.62)
CFR12RRB183--	.018	100	F, G, J, K	CFR16RRA394--	.39	50	F, G, J, K	.095(2.41)	.395(10.03)	.395(10.03)	.300(7.62)
CFR12RRH682--	.0068	150	F, G, J, K	CFR16RRA474--	.47	50	F, G, J, K	.145(3.68)	.395(10.03)	.395(10.03)	.300(7.62)
CFR12RRH822--	.0082	150	F, G, J, K	CFR16RRA564--	.56	50	F, G, J, K	.145(3.68)	.395(10.03)	.395(10.03)	.300(7.62)
CFR12RRC392--	.0039	200	F, G, J, K	CFR16RRA684--	.68	50	F, G, J, K	.195(4.95)	.395(10.03)	.395(10.03)	.300(7.62)
CFR12RRC472--	.0047	200	F, G, J, K	CFR16RRA824--	.82	50	F, G, J, K	.195(4.95)	.395(10.03)	.395(10.03)	.300(7.62)
CFR12RRC562--	.0056	200	F, G, J, K	CFR16RRA105--	1.00	50	F, G, J, K	.195(4.95)	.395(10.03)	.395(10.03)	.300(7.62)
CFR12RRK102--	.0010	250	F, G, J, K								
CFR12RRK122--	.0012	250	F, G, J, K								
CFR12RRK152--	.0015	250	F, G, J, K								
CFR12RRK182--	.0018	250	F, G, J, K								
CFR12RRK222--	.0022	250	F, G, J, K								
CFR12RRK272--	.0027	250	F, G, J, K								
CFR12RRK332--	.0033	250	F, G, J, K								

1/ APPLICATION NOTE. Care must be taken during wave soldering to assure that the CFR12 type capacitors are not damaged due to overheating. During the wave soldering operation, the CFR12 type capacitor shall not be exposed to conditions in excess of the resistance to solder heat requirements of MIL-C-55514.

2/ The complete type designation will include additional capacitance tolerance and failure rate level (M, P, R, or S).

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TABLE II. Substitutability data.

From		To	From		To
MIL-C-55514/8(ER) and MIL-C-55514/8A	MIL-C-55514/8B and MIL-C-55514/8C	MIL-C-55514/8D	MIL-C-55514/8(ER) and MIL-C-55514/8A	MIL-C-55514/8B and MIL-C-55514/8C	MIL-C-55514/8D
Type designation 1/	Type designation 1/	Type designation 1/	Type designation 1/	Type designation 1/	Type designation 1/
CFR12RRJ102--	CFR12RRK102--	CFR12RRK102--	CFR12RRG102--	CFR12RRK102--	CFR12RRK102--
CFR12RRJ122--	CFR12RRK122--	CFR12RRK122--	CFR12RRG122--	CFR12RRK122--	CFR12RRK122--
CFR12RRJ182--	CFR12RRK182--	CFR12RRK182--	CFR12RRG182--	CFR12RRK182--	CFR12RRK182--
CFR12RRJ222--	CFR12RRK222--	CFR12RRK222--	CFR12RRG222--	CFR12RRK222--	CFR12RRK222--
CFR12RRJ272--	CFR12RRK272--	CFR12RRK272--	CFR12RRG272--	CFR12RRK272--	CFR12RRK272--
CFR12RRJ332--	CFR12RRK332--	CFR12RRK332--	CFR12RRG332--	CFR12RRK332--	CFR12RRK332--
CFR12RRJ392--	CFR12RRC392--	CFR12RRC392--	CFR12RRG392--	CFR12RRC392--	CFR12RRC392--
CFR12RRJ472--	CFR12RRC472--	CFR12RRC473--	CFR12RRG472--	CFR12RRC472--	CFR12RRC472--
CFR12RRJ562--	CFR12RRC562--	CFR12RRC562--	CFR12RRG562--	CFR12RRC562--	CFR12RRC562--
CFR12RRJ682--	CFR12RRH682--	CFR12RRH682--	CFR12RRG682--	CFR12RRH682--	CFR12RRH682--
CFR12RRJ822--	CFR12RRH822--	CFR12RRH822--	CFR12RRG822--	CFR12RRH822--	CFR12RRH822--
CFR12RRJ103--	CFR12RRB103--	CFR12RRB103--	CFR12RRG103--	CFR12RRB103--	CFR12RRB103--
CFR12RRJ123--	CFR12RRB123--	CFR12RRB123--	CFR12RRG123--	CFR12RRB123--	CFR12RRB123--
CFR12RRJ153--	CFR12RRB153--	CFR12RRB153--	CFR12RRG153--	CFR12RRB153--	CFR12RRB153--
CFR12RRJ183--	CFR12RRB183--	CFR12RRB183--	CFR12RRG183--	CFR12RRB183--	CFR12RRB183--
CFR12RRJ223--	CFR12RRG223--	CFR12RRG223--	CFR12RRB102--	CFR12RRK102--	CFR12RRK102--
CFR12RRJ273--	CFR12RRB273--	CFR12RRB273--	CFR12RRB122--	CFR12RRK122--	CFR12RRK122--
CFR12RRJ333--	CFR12RRA333--	CFR12RRA333--	CFR12RRB182--	CFR12RRK182--	CFR12RRK182--
CFR12RRJ393--	CFR12RRA393--	CFR12RRA393--	CFR12RRB222--	CFR12RRK222--	CFR12RRK222--
CFR12RRJ473--	CFR12RRA473--	CFR12RRA473--	CFR12RRB272--	CFR12RRK272--	CFR12RRK272--
CFR12RRJ563--	CFR12RRA563--	CFR12RRA563--	CFR12RRB332--	CFR12RRK332--	CFR12RRK332--
CFR12RRJ683--	CFR12RRA683--	CFR16RRA683--	CFR12RRB392--	CFR12RRC392--	CFR12RRC392--
CFR12RRJ823--	CFR12RRA823--	CFR16RRA823--	CFR12RRB472--	CFR12RRC472--	CFR12RRC472--
CFR12RRJ104--	CFR16RRA104--	CFR16RRA104--	CFR12RRB562--	CFR12RRC562--	CFR12RRC562--
CFR12RRA102--	CFR12RRK102--	CFR12RRK102--	CFR12RRB682--	CFR12RRH682--	CFR12RRH682--
CFR12RRA122--	CFR12RRK122--	CFR12RRK122--	CFR12RRB822--	CFR12RRH822--	CFR12RRH822--
CFR12RRA182--	CFR12RRK182--	CFR12RRK182--	CFR12RRH102--	CFR12RRK102--	CFR12RRK102--
CFR12RRA222--	CFR12RRK222--	CFR12RRK222--	CFR12RRH122--	CFR12RRK122--	CFR12RRK122--
CFR12RRA272--	CFR12RRK272--	CFR12RRK272--	CFR12RRH182--	CFR12RRK182--	CFR12RRK182--
CFR12RRA332--	CFR12RRK332--	CFR12RRK332--	CFR12RRH222--	CFR12RRK222--	CFR12RRK222--
CFR12RRA392--	CFR12RRC392--	CFR12RRC392--	CFR12RRH272--	CFR12RRK272--	CFR12RRK272--
CFR12RRA472--	CFR12RRC472--	CFR12RRC472--	CFR12RRH332--	CFR12RRK332--	CFR12RRK332--
CFR12RRA562--	CFR12RRC562--	CFR12RRC562--	CFR12RRH392--	CFR12RRC392--	CFR12RRC392--
CFR12RRA682--	CFR12RRH682--	CFR12RRH682--	CFR12RRH472--	CFR12RRC472--	CFR12RRC472--
CFR12RRA822--	CFR12RRH822--	CFR12RRH822--	CFR12RRH562--	CFR12RRC562--	CFR12RRC562--
CFR12RRA103--	CFR12RRB103--	CFR12RRB103--	CFR12RRC102--	CFR12RRK102--	CFR12RRK102--
CFR12RRA123--	CFR12RRB123--	CFR12RRB123--	CFR12RRC122--	CFR12RRK122--	CFR12RRK122--
CFR12RRA153--	CFR12RRB153--	CFR12RRB153--	CFR12RRC182--	CFR12RRK182--	CFR12RRK182--
CFR12RRA183--	CFR12RRB183--	CFR12RRB183--	CFR12RRC222--	CFR12RRK222--	CFR12RRK222--
CFR12RRA223--	CFR12RRG223--	CFR12RRG223--	CFR12RRC272--	CFR12RRK272--	CFR12RRK272--
CFR12RRA273--	CFR12RRG273--	CFR12RRG273--	CFR12RRC332--	CFR12RRK332--	CFR12RRK332--

1/ The complete type designation will include additional capacitance tolerance and failure rate level (M, P, R, or S).

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Substitutability for capacitance tolerance shall be as follows:

Cap. tolerance	Will replace cap. tolerance
F	G, J, K
G	J, K
J	K
K	

Substitutability for voltage shall be as follows:

Volts dc	Will replace voltages
250	200, 150, 100, 75, 50
200	150, 100, 75, 50
150	100, 75, 50
100	75, 50
75	50
50	

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - OS
Air Force - 17, 99
DLA - ES

User activities:
Navy - AS, MC
Air Force -19

Preparing activity:
Army - ER

Agent:
DLA - ES

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